

Assignment 4

Choose at least one of the following assignments.

a) Sepsis prediction

Select one of the following models: **GRU-ODE**, **ODE-RNN**, or **ODE-LSTM**, and apply it to the Sepsis dataset. The objective is the early detection of sepsis using clinical ICU data. Specifically, your model must predict whether a patient will develop sepsis within a 6-hour window prior to clinical onset. Submit your code to Whiteboard by 02.12.2025 and present it in class.

Sepsis dataset

The **B/time_series** directory contains 20,000 individual files (20,000 time series), each representing a unique patient's ICU visit. Within these files, every row corresponds to a single hour of data, capturing 40 recorded clinical variables.

1. **Vital Signs (8 columns):** High-frequency measurements like Heart Rate (HR), Pulse Oximetry (O2Sat), Temperature (Temp), Systolic/Mean/Diastolic Blood Pressure (SBP, MAP, DBP), Respiration Rate (Resp), and End-tidal CO2 (EtCO2).
2. **Laboratory Values (26 columns):** Blood tests that are measured less frequently, such as Lactate, White Blood Cell count (WBC), Creatinine, Platelets, Glucose, Bilirubin, etc.
3. **Demographics (6 columns):** Static features like Age, Gender, Hospital Admit Time, and ICU Length of Stay (ICULOS).

B/labels.csv contains binary classification labels corresponding to each time series file. A label of 0 indicates the patient did not develop sepsis within 6 hours following the end of the recorded time series, while a label of 1 indicates the patient developed sepsis within that 6-hour window.

More information about the data can be found in <https://physionet.org/content/challenge-2019/1.0.0/>.

This is the whole dataset, you have to split it into train, validation, and test sets yourself.

b) Open source contribution

Choose one of **GRU-ODE**, **ODE-RNN**, **ODE-LSTM** and create a Jupyter notebook demonstrating how to implement the chosen model using **Difffrax**, **Equinox**, and **Optax**. Make sure your code has a similar style to the ones in <https://github.com/patrick-kidger/diffrax/tree/main/docs/examples>. You can use the Spirals dataset from Assignment 3. Submit your notebook to Whiteboard by 02.12.2025 and present it in class. Prepare this as if it was a contribution, but don't actually submit it to **Difffrax** just yet.

c) Model implementation

Choose one of NCDE or NSDE models from Table 1 in <https://arxiv.org/abs/2502.09885>.

- If you choose an NCDE model: Apply to the Spirals task from Assignment 3.
- If you choose an NSDE model: Apply it to Covid-19 and Lynx-Hare tasks from Assignment 1.
- If you want to implement another Neural Differential equation based model just drop me an email at u.kaya@fu-berlin.de about it.

Submit your code to Whiteboard by 02.12.2025 and present it in class.